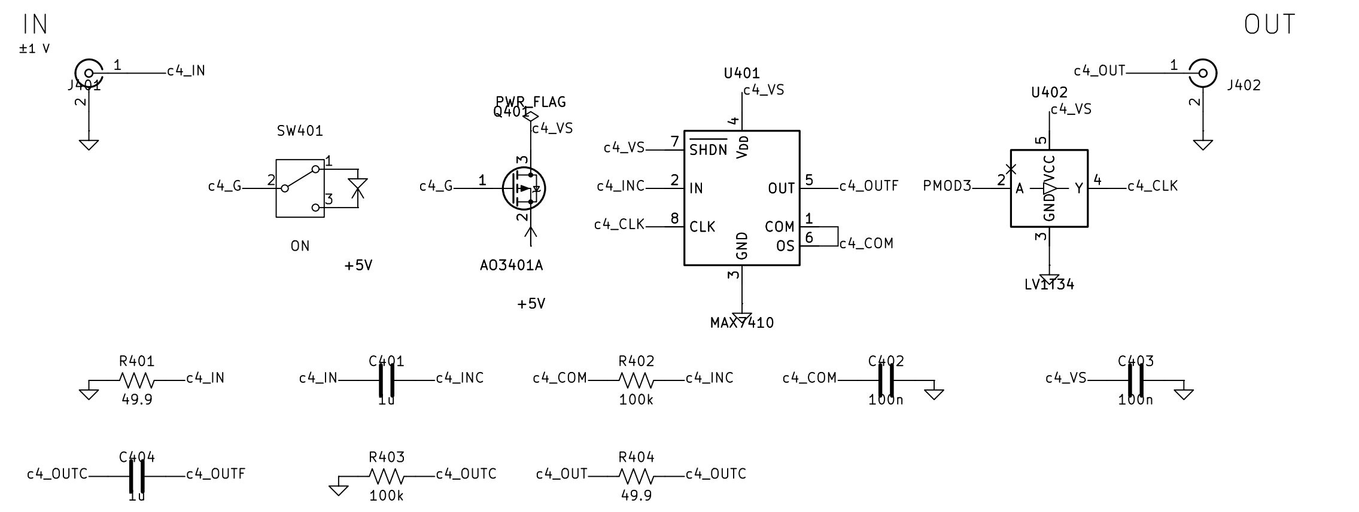
c4 SC filter, MAX7410

tutorial Switched capacitors (+5 V)

cell (0,0)

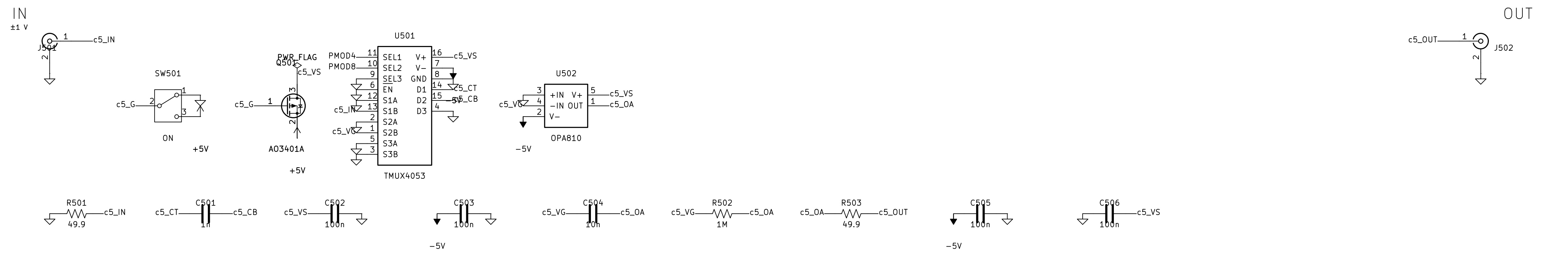

IN
±1 V
OUT
SW401
Q401
AO3401A
U401
MAX7410
U402
LV1T34
PWR_FLAG
+5V

EXPECT: a 5th-order Butterworth low-pass at fCLK / 100 (up to 15 kHz from a 1.5 MHz clock on IO3); the clock's images near fCLK
MAX7410: a switched-capacitor ladder whose corner is the clock divided by 100, clocked from IO3 through an SN74LV1T34.
IN and OUT sit at COM (2.5 V), so both are AC-coupled. Sweep the clock and watch the corner follow; alias a tone.

c5 SC integrator, TMUX4053

tutorial Switched capacitors (+-5 V)

cells (1,0), (2,0)

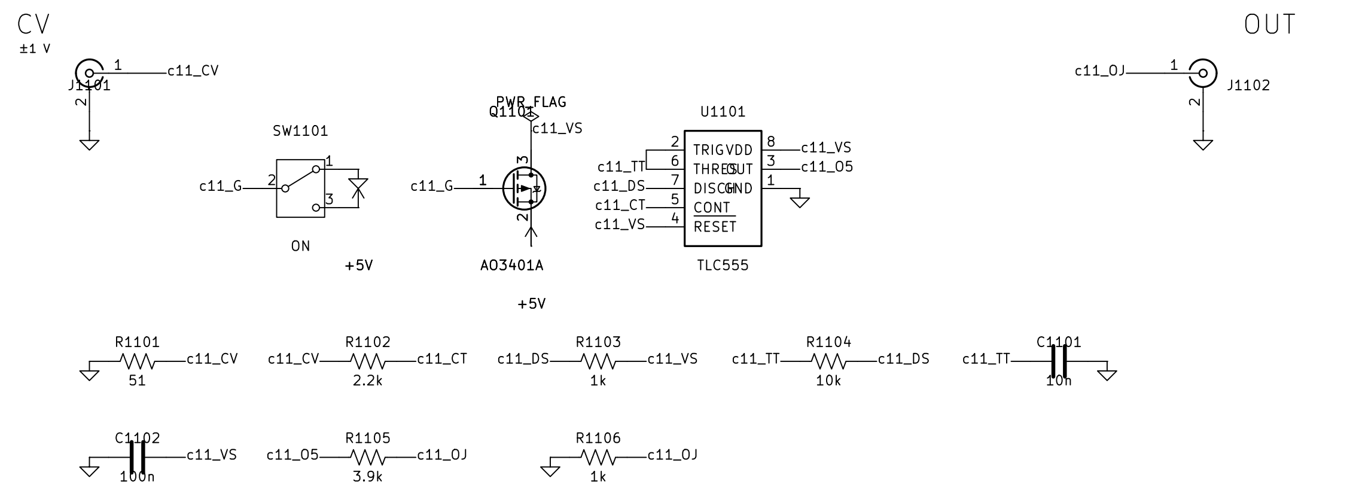

IN
±1 V
OUT
SW501
Q501
AO3401A
U501
TMUX4053
U502
OPA810
+5V
-5V

EXPECT: an integrator whose input resistor is 1 / (f x 1 nF): 1 M at 1 kHz, 1 k at 1 MHz; OUT = -(1 / R C) x the integral of IN
phi1 (IO4) connects the 1 nF's top plate to IN, phi2 (IO8) its bottom plate to the summing node: C x Vin per cycle at f,
a resistor of 1 / (f C). Make the phases non-overlapping, or overlap them and watch IN leak through. 1 M sets the DC gain.

c11 555, TLC555

tutorial Trading speed for bits (+5 V)

cell (3,0)

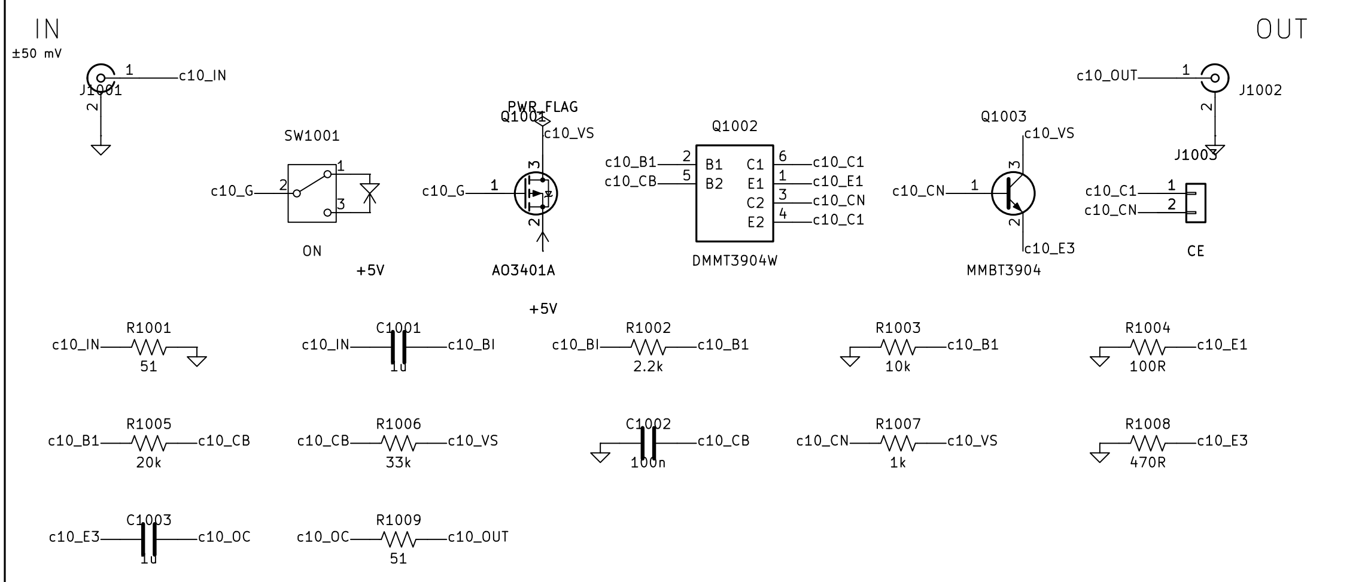

CV
±1 V
OUT
SW1101
Q1101
AO3401A
U1101
TLC555
+5V

EXPECT: a square near 6.9 kHz on OUT (5 V divided to about 1 V); CV moves it: up as CV falls, down as it rises
TLC555 astable: f = 1.44 / ((1 k + 2 x 10 k) x 10 nF), the capacitor between 1/3 and 2/3 of the supply. CV pulls the
upper threshold through 2.2 k: a voltage-to-frequency converter. Count OUT's cycles in the FPGA and you have an ADC.

c10 cascode

tutorial Transistors (+5 V)

cell (0,1)

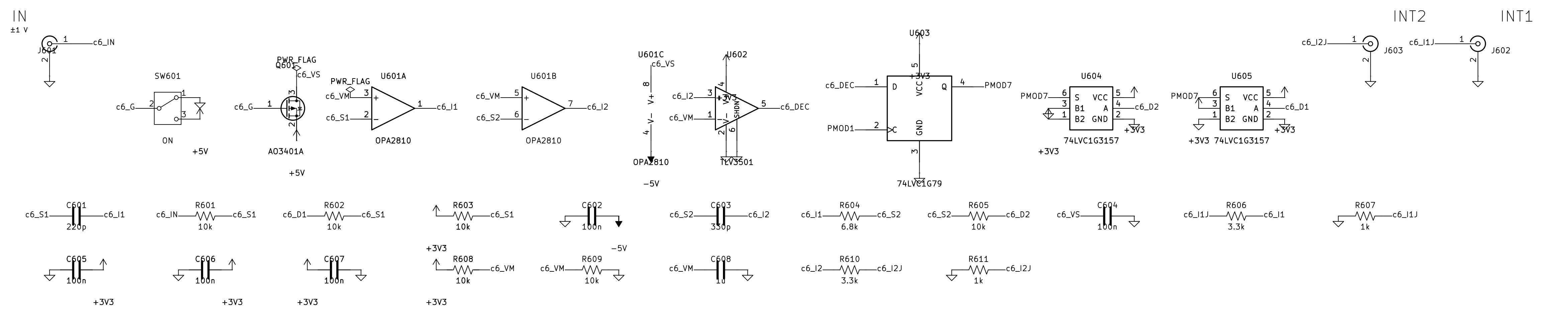

IN
±50 mV
OUT
SW1001
Q1001
AO3401A
Q1002
DMMT3904W
Q1003
MMBT3904
J1003
CE
+5V

EXPECT: gain about -6 into a high impedance (-8 at Q1's collector), half that into 50 ohm; common emitter (jumper on) about 2 MHz wide, the cascode (jumper parked) several times wider
Q1's collector-base capacitance, multiplied by the gain (Miller), meets the 2.2 k at its base; the cascode holds Q1's
collector still, so only one Ccb counts. The jumper on the header shorts the cascode out: compare the two sweeps.

c6 delta-sigma, second order

tutorial Trading speed for bits (+-5 V)

cells (1,1), (2,1)

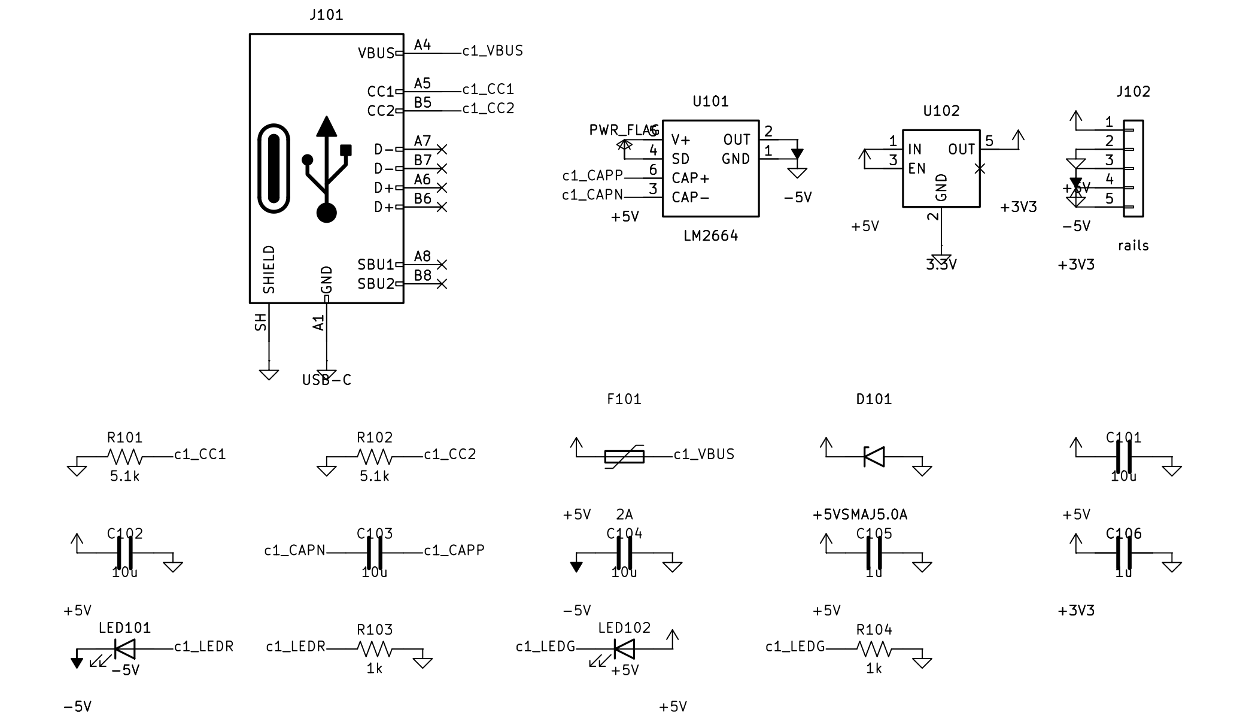

IN
±1 V
INT2
INT1
SW601
Q601
AO3401A
U601A
OPA2810
U601B
OPA2810
U601C
U602
TLV3501
U603
74LVC1G79
U604
74LVC1G3157
U605
74LVC1G3157
+5V
-5V
+3V3

EXPECT: a one-bit stream on IO7 whose density follows IN (+-1.65 V full scale), its noise shaped at 40 dB per decade
Clock IO1 at about 1 MHz and count IO7's ones. The second integrator pushes the quantization noise up at 40 dB per
decade (AMOD-A's a6 is first order: 20 dB), so a sinc3 decimator gets more bits per clock. INT1, INT2: high impedance.

c1 power, USB-C 5 V in: +5, -5, +3.3 out

tutorial Regulators, Switched capacitors

cell (3,1)

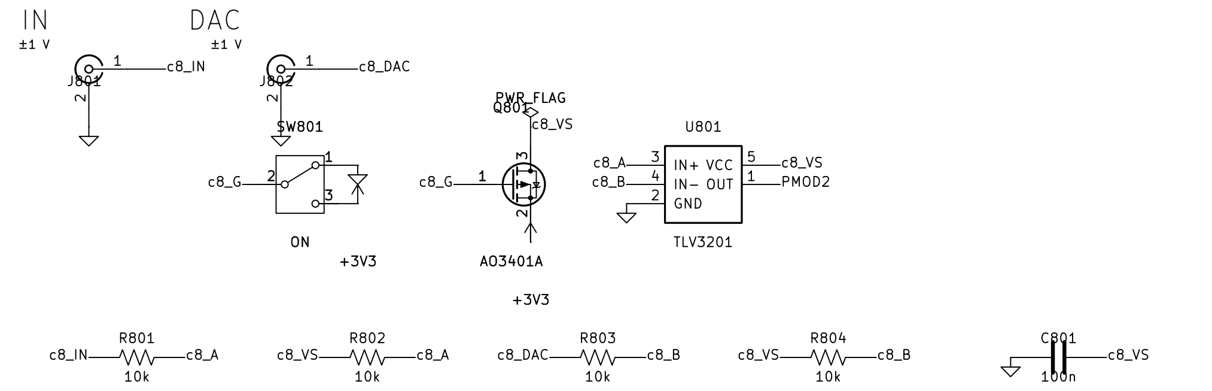

J101
USB-C
U101
LM2664
U102
J102
rails
F101
D101
LED101
LED102
+5V
-5V
+3V3

EXPECT: +5 V from the USB-C (up to 1.5 A), -5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that -5 V is up
USB-C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes -5 V for the op-amp tiles (40 mA: four tiles at 10 mA); it is not switched per tile, an op
amp with its +5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, -5, ground, +3.3: the bench's rails for a breadboard experiment.

c8 comparator, TLV3201

tutorial Trading speed for bits (+3.3 V)

cell (0,2)

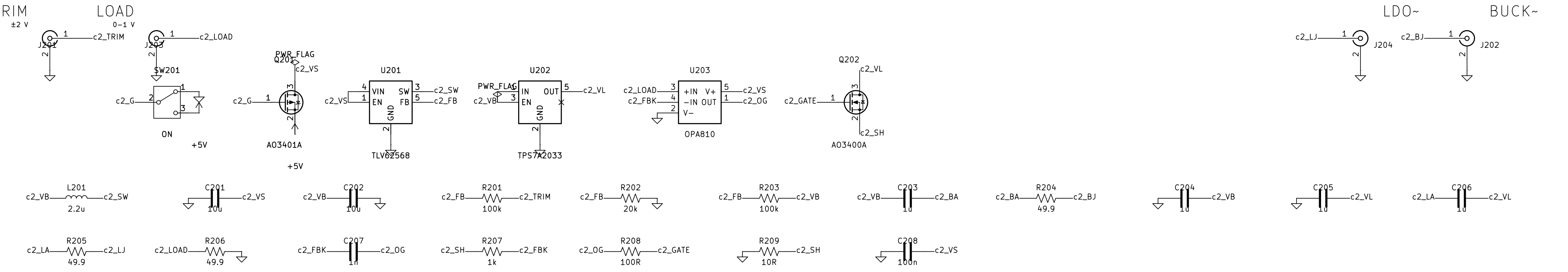

IN
±1 V
DAC
SW801
Q801
AO3401A
U801
TLV3201
+3V3

EXPECT: BIT on Pmod IO2 is high while IN is above DAC; a SAR search in gateware gives 12 bits in 12 steps of the DAC
TLV3201 (40 ns, 1.2 mV hysteresis) on the 3.3 V rail. Each input is averaged with the rail through a matched pair
of 0.1 % 10 k, so +-1 V lands at 1.15-2.15 V on both pins and the comparison is unchanged. 20 k input impedance:
drive it from a 50 ohm source, read the bit on IO2. A successive-approximation ADC: the gateware sets the host's
DAC, reads the bit, halves the step; a tracking ADC: up when high, down when low; a slope ADC with a ramp on DAC.

c2 regulators, buck, LDO, load

tutorial Regulators (+5 V)

cells (1,2), (2,2)

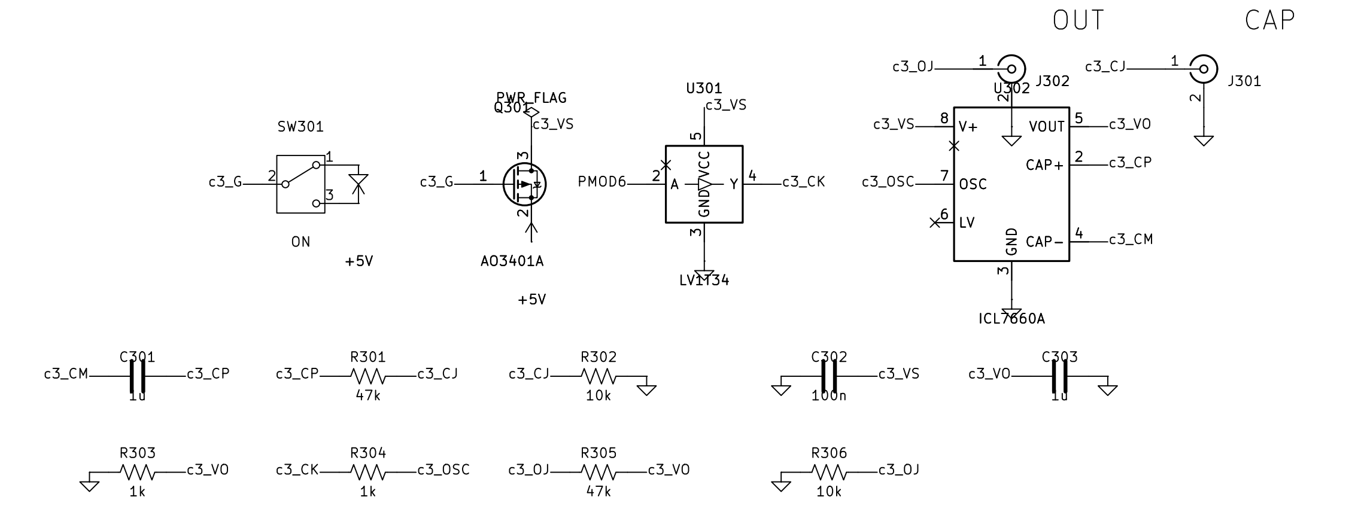

TRIM
±2 V
LOAD
0-1 V
LDO~
BUCK~
SW201
Q201
AO3401A
U201
TLV62568
U202
TPS7A2033
U203
OPA810
Q202
AO3400A
+5V

EXPECT: BUCK~ shows the buck's 1.5 MHz ripple (a few mV, pulse-skipping below about 150 mA); LDO~ the same after the LDO, 40 dB smaller; LOAD sets 100 mA per volt
TLV62568: 3.6 V from 100 k / 20 k; TRIM through 100 k into the feedback node moves it (0 V: 4.2 V, +1 V: 3.2 V, open: 3.6 V).
TPS7A2033: 3.3 V, about 92 dB of PSRR at 1 kHz and 40 dB at 1 MHz; below 3.5 V in it drops out (lower TRIM's voltage).
The load: an OPA810 servos 10 ohm to LOAD's voltage through an AO3400A (1 k and 1 nF slow its loop): a square on LOAD
is a load step. Keep it under 100 mA for long (the shunt is 1/8 W), steps can be larger. Both outputs are AC-coupled.

c3 charge pump, ICL7660A

tutorial Regulators (+5 V)

cell (3,2)

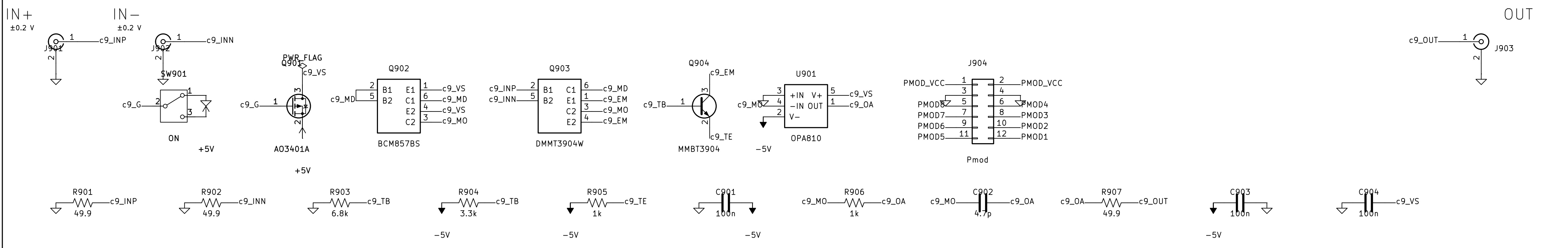

OUT
CAP
SW301
Q301
AO3401A
U301
LV1T34
U302
ICL7660A
+5V

EXPECT: about -5 V on the output, falling with load as 1 / (f C) + the switches: 2 k at 1 kHz, 80 ohm at 100 kHz (the pump at half IO6's clock)
ICL7660A: switches charge the flying 1 uF across the supply and dump it, upside down, into the output's 1 uF. IO6
clocks it through an SN74LV1T34 and 1 k; with 1 k of load, sweep the clock and read OUT: Rout = 1 / (f C) + 60 ohm.

c9 differential pair

tutorial Transistors (+-5 V)

cells (0,3), (1,3)

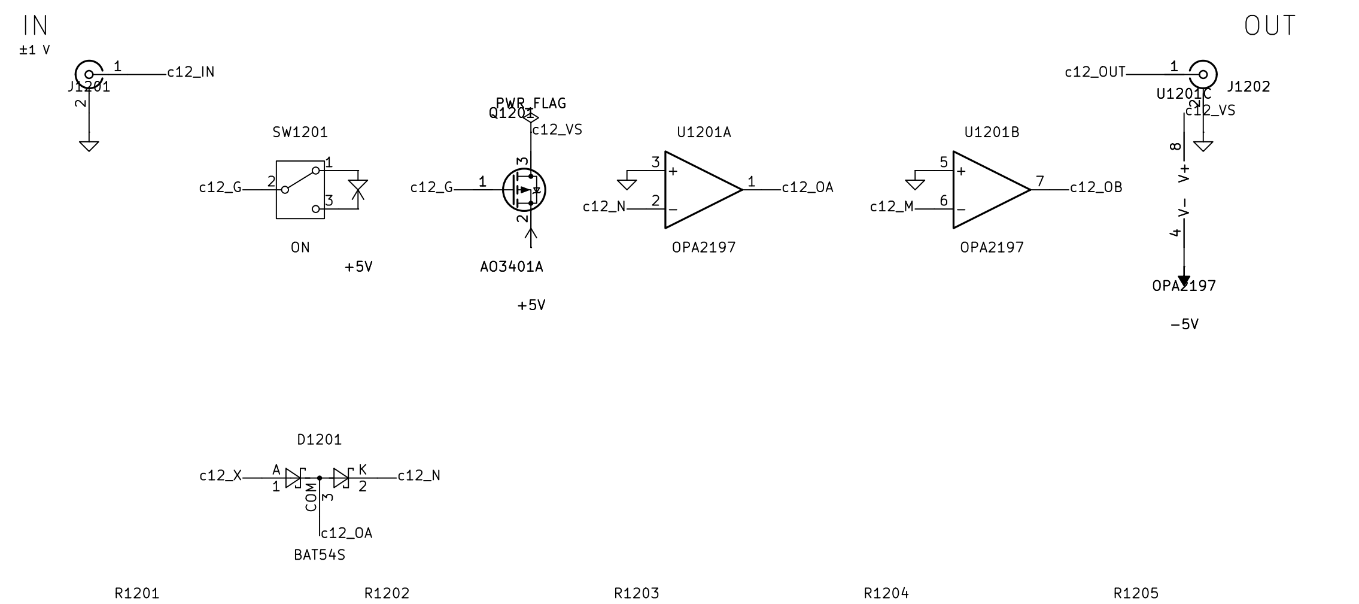

IN+
±0.2 V
IN-
±0.2 V
OUT
SW901
Q901
AO3401A
Q902
BCM857BS
Q903
DMMT3904W
Q904
MMBT3904
U901
OPA810
J904
Pmod
+5V
-5V

EXPECT: OUT = -1 k x (Ic1 - Ic2): +-1 V, tanh of (IN+ - IN-) / 2 VT, linear within about +-25 mV of balance
A matched NPN pair (DMMT3904W: 2 mV of VBE, 2 % of hFE) on a 1 mA tail; a matched PNP pair as the mirror; an OPA810
holds the output node at 0 V and turns the difference current into volts. Sweep IN+ with IN- at 0 V: the tanh, its
slope gm = Itail / (4 VT), about 10 mS here; a triangle in becomes a sine-like wave out when it is driven past balance.

c12 rectifier, precision

tutorial Transistors (+-5 V)

cell (2,3)

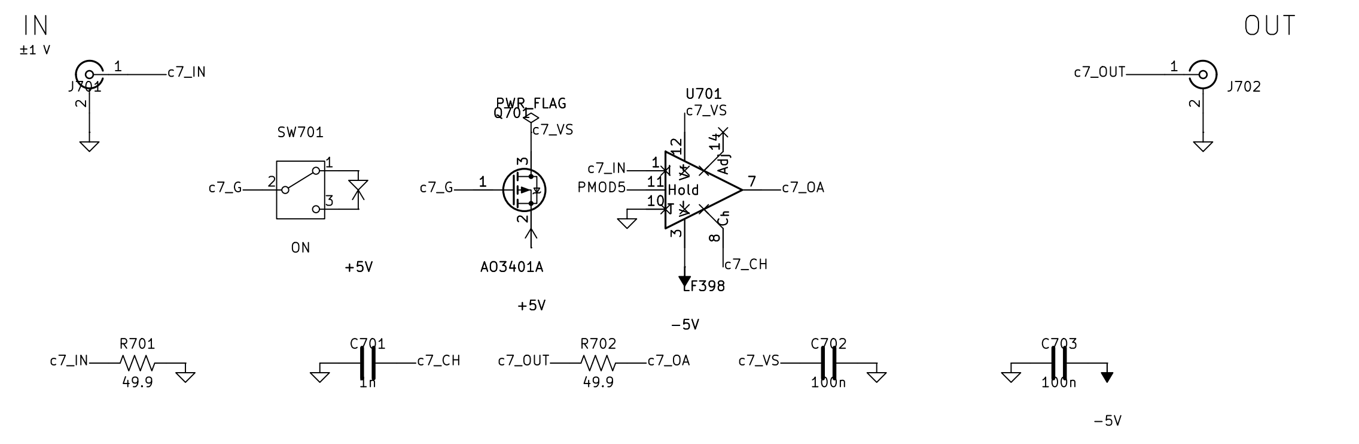

IN
±1 V
OUT
SW1201
Q1201
AO3401A
U1201A
OPA2197
U1201B
OPA2197
D1201
BAT54S
+5V
-5V

EXPECT: OUT = |IN| from DC to about 100 kHz; above that the first stage's slew across the diodes leaves a glitch at each zero crossing
Stage 1: inverting with BAT54S in its feedback, so its node X follows -IN for positive IN and stays at 0 for negative.
Stage 2: OUT = -(IN + 2 X) = |IN|. No dead zone from the diodes (the op amp's gain hides their drop) until the op amp
must slew through two diode drops at each zero crossing: watch the glitch grow with frequency.

c7 track and hold, LF398

tutorial Trading speed for bits (+-5 V)

cell (3,3)


IN
OUT
SW701
Q701
AO3401A
U701
LF398
+5V
-5V

EXPECT: OUT follows IN while IO5 is high and holds when it falls: 4 us to acquire, about 30 mV/s of droop, a few mV of hold step
LF398 on +-5 V (its minimum supply: inputs within +-1.5 V). 1 nF C0G: acquisition about 4 us, droop about 30 mV/s.
Measure the acquisition time, the droop, the pedestal (the hold step) against IN, the feedthrough of a sine while
holding, and the aperture delay against the FPGA's edge on IO5.

no mounting holes: four stick-on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/amod_c.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a buck and an LDO under load, a charge pump, switched capacitors, a second-order delta-sigma, sa
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/
Sheet: /
File: amod_c.kicad_sch
Title: AMOD-C: power and data conversion
Size: A1
Date: 2026-10-25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1